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Method to embed planar FETs with finFETs

Abstract

Various embodiments of the present disclosure are directed towards a method to embed planar field-effect transistor (FETs) with fin field-effect transistors (finFETs). A semiconductor substrate is patterned to define a mesa and a fin. A trench isolation structure is formed overlying the semiconductor substrate and surrounding the mesa and the fin. A first gate dielectric layer is formed on the mesa, but not the fin. The trench isolation structure recessed around the fin, but not the mesa, after the forming the first gate dielectric layer. A second gate dielectric layer is deposited overlying the first gate dielectric layer at the mesa and further overlying the fin. A first gate electrode is formed overlying the first and second gate dielectric layers at the mesa and partially defining a planar FET. A second gate electrode is formed overlying the second gate dielectric layer at the fin and partially defining a finFET.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS (1) This Application is a Continuation of U.S. application Ser. No. 18/365,424, filed on Aug. 4, 2023, which is a Continuation of U.S. application Ser. No. 17/751,958, filed on May 24, 2022 (now U.S. Pat. No. 11,830,875, issued on Nov. 28, 2023), which is a Divisional of U.S. application Ser. No. 16/858,801, filed on Apr. 27, 2020 (now U.S. Pat. No. 11,355,493, issued on Jun. 7, 2022), which claims the benefit of U.S. Provisional Application No. 62/988,967, filed on Mar. 13, 2020. The contents of the above-referenced Patent Applications are hereby incorporated by reference in their entirety.

BACKGROUND

(1) The integrated circuit (IC) manufacturing industry has experienced exponential growth over the last few decades. As ICs have evolved, functional density (e.g., the number of interconnected devices per chip area) has increased while feature sizes have decreased. One advancement by the IC manufacturing industry to increase functional density and decrease feature sizes is the development of fin field-effect transistors (finFETs). While finFETs have several advantages over traditional planar FETs (e.g., reduced power consumption, smaller feature sizes, etc.), the use of finFETs does not come without problems.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIGS. 1A-1D respectively illustrate top and cross-sectional views of some embodiments of an IC comprising a planar field-effect transistor (FET) and a fin FET (finFET).
- (3) FIG. 2 is a perspective view of some embodiments of the IC of FIGS. 1A-1D.
- (4) FIGS. 3A-3E illustrate cross-sectional views of some alternative embodiments of the IC of FIG. 1B.
- (5) FIGS. 4A and 4B illustrate cross-sectional views of some alternative embodiments of the IC of

FIG. 1C.

(6) FIG. 5 illustrates a cross-sectional view of some embodiments of an IC comprising a planar FET and a finFET that are spaced from each other in a direction extending along corresponding channel lengths.

(7) FIGS. 6A and 6B illustrate cross-sectional views of some alternative embodiments of the IC of FIG. 5.

(8) FIGS. 7A and 7B illustrate cross-sectional views of some embodiments of the IC of FIG. 5 in which the IC further comprises an interconnect structure.

(9) FIGS. 8A and 8B respectively illustrate top and cross-sectional views of some embodiments of an IC comprising a planar FET and a plurality of finFETs that are spaced from each other in a direction extending transverse to corresponding channel lengths.

(10) FIGS. 9A and 9B respectively illustrate top and cross-sectional views of some embodiments of an IC comprising a plurality of planar FETs and a plurality of finFETs that are electrically coupled to an interconnect structure.

(11) FIGS. 10A and 10B through FIGS. 13A and 13B, FIGS. 14A-14C through FIGS. 23A-23C, FIGS. 24A and 24B, FIGS. 25A-25D through FIGS. 28A-28D, FIGS. 29A and 29B through FIGS. 37A and 37B illustrate a series of views of some embodiments of a method for forming an IC comprising a planar FET and a finFET.

(12) FIG. 38 illustrates a block diagram of some embodiments of a method for forming an IC comprising a planar FET and a finFET.

DETAILED DESCRIPTION

(13) The present disclosure provides many different embodiments, or examples, for implementing different features of this disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(14) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(15) An integrated circuit (IC) may, for example, comprise a plurality of fin field-effect transistors (finFETs) that define a logic core and input/output (IO). A challenge with the IC is that IO finFETs may be configured for low operating voltages. For example, the IO finFETs may be configured to operate at 1.8 volts or some other suitable voltage. Due to the low operating voltages, the IO finFETs may be unsuitable for high voltage (HV) applications, non-volatile memory (NVM) applications, and other suitable applications.

(16) A solution is to reconfigure the IO finFETs for higher operating voltages. For example, the IO finFETs may be reconfigured to operate at 2.5 volts or some other suitable voltage. However, this may come at the expense of low voltage IO, which may still have application within the IC. Further, the IO finFETs may have short lifespans due to the higher operating voltages. Another solution is to integrate additional IO configured for the higher operating voltages into the IC while maintaining IO configured for the low operating voltages. However, this may pose process

challenges as discussed hereafter. Further, the additional IO finFETs configured for the higher operating voltages may still have short lifespans.

(17) According to a method for integrating low voltage IO with high voltage IO, fins are defined in a semiconductor substrate. A shallow trench isolation (STI) structure is formed surrounding the fins, and dopants are implanted into the fins to define wells for finFETs being formed. The STI structure is recessed around the fins, and dopants are again implanted into the fins to tune threshold voltages and/or other suitable properties of the finFETs being formed. A high voltage IO dielectric layer is deposited on the fins and is subsequently removed from the fins corresponding to the low voltage IO. A low voltage IO dielectric layer is deposited on the fins and polysilicon gate electrodes are formed over the high and low IO dielectric layers.

(18) A challenge is that formation of the high and/or low voltage IO dielectric layer(s) after the tuning may change properties of the finFETs. For example, the high and low voltage IO dielectric layers may be formed by thermal oxidation for high quality and low leakage. However, thermal oxidation may partially consume the fins and may hence erode dopants implanted to tune threshold voltages and/or other suitable properties of the finFETs. This may, in turn, change the threshold voltages and/or other suitable properties of the finFETs.

(19) Methods for forming the IC with low voltage IO, but without high voltage IO, may be designed to compensate for the changes in doping profile during formation of the low voltage IO dielectric layer. However, when high voltage IO is integrated into these methods as described above, there may be no compensation for the changes in doping profile during formation of the high voltage IO dielectric layer. As a result, manufacturing yields may be negatively impacted without costly rework of the methods.

(20) Various embodiments of the present disclosure are directed towards a method for forming a planar FET and a finFET together, as well as an IC resulting from the method. According to some embodiments of the method, a semiconductor substrate is patterned to define a mesa and a fin. A trench isolation structure is formed surrounding the mesa and the fin. A first dielectric layer is deposited on the mesa and the fin and is then removed from the fin. The trench isolation structure is recessed after removing the first dielectric layer from the fin, and dopants are implanted into the semiconductor substrate (e.g., at the fin and/or some other suitable location(s)) to tune threshold voltages and/or other suitable properties for semiconductor devices being formed. A second dielectric layer is deposited overlying the first dielectric layer at the mesa and further overlying the fin. A first gate electrode is formed overlying the first and second dielectric layers at the mesa and partially defining a planar FET. Further, a second gate electrode is formed overlying the second dielectric layer at the fin and partially defining a finFET.

(21) The planar FET may be employed for IO, whereas the finFET may be employed for lower voltage IO or a logic core. For example, the planar FET may be employed for 2.5-volt IO, whereas the finFET may be employed for 1.8-volt IO. Other suitable voltages are, however, amenable. FinFETs may have reduced power consumption, smaller feature sizes, and other benefits over planar FETs. However, finFETs may be less reliable than planar FETs. Hence, IO finFETs may have short lifespans when operated at higher voltages. By using the planar FET for higher voltage IO while using the finFET for a logic core or lower voltage IO, the lifespan of the IC may be enhanced. The planar FET mitigates reliability issues, and the finFET provides the aforementioned benefits where reliability is less of an issue. Further, because the planar FET may be employed for higher voltage IO, the IC may be used for HV applications, NVM applications, and other suitable applications that depend upon higher voltages.

(22) By forming the first dielectric layer before the recessing, the first gate dielectric layer may be formed without negatively impacting doping profiles of semiconductor devices on the semiconductor substrate. For example, the first dielectric layer may be formed by thermal oxidation for high quality and low leakage. Thermal oxidation may consume the semiconductor substrate. If the first dielectric layer was formed after the recessing and after the tuning, thermal oxidation may

consume portions of the semiconductor substrate at which dopants are implanted for the tuning. This, in turn, may change doping profiles of the semiconductor devices and may hence change the threshold voltages and/or other suitable parameters. Because the first dielectric layer may be formed without negatively impacting doping profiles of the semiconductor devices, the first dielectric layer and hence the planar FET may be formed without negatively impacting yields for the semiconductor devices. The semiconductor devices may include, for example, static random-access memory (SRAM) devices and/or other suitable semiconductor device(s). Further, the first dielectric layer and hence the planar FET may be integrated with and formed with no impact or minimal impact on existing finFET processes.

(23) With reference to FIGS. 1A-1D, top and cross-sectional views **100A-100D** of some embodiments of an IC comprising a planar FET **102** and a plurality of finFETs **104** is provided. FIG. 1A is a top view **100A** of the IC, whereas FIGS. 1B-1D are cross-sectional views **100B-100D** respectively along lines A-A', B-B', and C-C' in FIG. 1A. The plurality of finFETs **104** comprises a core finFET **104c** and an IO finFET **104io**. The planar FET **102** is employed for IO, and the core and IO finFETs **104c**, **104io** are respectively employed for a logic core and lower voltage IO. For example, the planar FET **102** may be employed for 2.5-volt IO, and the IO finFET **104io** may be employed for 1.8-volt IO. Other suitable voltages are, however, amenable.

(24) FinFETs may have reduced power consumption, smaller feature sizes, and other benefits over planar FETs. However, finFETs may be less reliable than planar FETs. Hence, IO finFETs may have shorter lifespans when operated at higher voltages. By using the planar FET **102** for higher voltage IO while using the IO finFET **104io** for lower voltage IO, the lifespan of the IC may be enhanced. The planar FET **102** mitigates reliability issues from higher voltages, and the IO finFET **104io** provides the aforementioned benefits where reliability is less of an issue. Further, because the planar FET **102** may be employed for higher voltage IO, the IC may be used for HV applications, NVM applications, and other suitable applications.

(25) The planar FET **102** and the finFETs **104** overlie a semiconductor substrate **106** and comprise corresponding gate electrodes **108** and corresponding gate dielectric layers **110**. The planar FET **102** is on a mesa **106m** defined by the semiconductor substrate **106**, and the finFETs **104** are respectively on fins **106f** defined by the semiconductor substrate **106**. The semiconductor substrate **106** may, for example, be or comprise a bulk substrate of monocrystalline silicon or some other suitable semiconductor substrate.

(26) The mesa **106m** has a length $L_{sub.m}$ and the fins **106f** have a length $L_{sub.f}$. In some embodiments, a ratio of the mesa length $L_{sub.m}$ to the fin length $L_{sub.f}$ is greater than or equal to about 1:1, 2:1, or 5:1. Other suitable values are, however, amenable. In some embodiments, the mesa **106m** has a width $W_{sub.m}$ that is greater than about 150 nanometers, but other suitable values are amenable. In some embodiments, the fins **106f** have individual widths $W_{sub.f}$ that are greater than about 16 nanometers or some other suitable values. In some embodiments, a top surface area of the mesa **106m** is an order of magnitude greater than individual top surface areas of the fins **106f**, but other suitable relationships between the top surface areas are amenable.

(27) The gate electrodes **108** respectively overlie the mesa **106m** and the fins **106f**. The gate electrode **108** of the planar FET **102** has a bottommost surface wholly or substantially elevated above a topmost surface of the mesa **106m**. In some embodiments, the gate electrode **108** of the planar FET **102** has a bottom with an inverted U-shaped profile, a flat or planar profile, or some other suitable profile. In some embodiments, the gate electrode **108** of the planar FET **102** has a bottommost point or edge elevated above a topmost point or edge of the mesa **106m**. The gate electrodes **108** of the finFETs **104** respectively wrap around tops of the fins **106f** and have bottom surfaces recessed relative to top surfaces of the fins **106f**. In some embodiments, the gate electrodes **108** of the finFETs **104** have bottoms with inverted U-shaped profiles or some other suitable profiles. The gate electrodes **108** may, for example, be or comprise metal, doped polysilicon, some other suitable conductive material(s), or any combination of the foregoing.

- (28) In some embodiments, the gate electrode **108** of the planar FET **102** has a length $L_{sub.g}$ greater than about 0.24 micrometers, 0.4 micrometers, 0.5 micrometers, 0.72 micrometers, 1.0 micrometers, or some other suitable value. Increasing the length $L_{sub.g}$ increases the voltage with which the planar FET **102** can sustain operation and/or increases the lifetime of the planar FET **102**. For example, increasing the length $L_{sub.g}$ from about 0.24 micrometers to about 1.0 micrometers may increase the lifetime of the planar FET **102** by about 2-3 orders of magnitude.
- (29) The gate dielectric layers **110** respectively separate the gate electrodes **108** from the semiconductor substrate **106**. The gate dielectric layer **110** of the planar FET **102** separates the gate electrode **108** of the planar FET **102** from the mesa **106m**, and the gate dielectric layers **110** of the finFETs **104** respectively separate the gate electrodes **108** of the finFETs **104** from the fins **106f**. The gate dielectric layers **110** have different thicknesses and/or material compositions to account for different operating conditions and/or usages. For example, the gate dielectric layer **110** of the planar FET **102** may be thicker than the gate dielectric layer **110** of the IO finFET **104io** because the planar FET **102** may be employed for higher voltage IO than the IO finFET **104io**.
- (30) In some embodiments, the gate dielectric layer **110** of the planar FET **102** is defined by a first IO dielectric layer **112**, a second IO dielectric layer **114**, and a core dielectric layer **116**, whereas the gate dielectric layer **110** of the IO finFET **104io** is defined by the second IO dielectric layer **114** and the core dielectric layer **116** but not the first IO dielectric layer **112**. In alternative embodiments, the core dielectric layer **116** is omitted from the gate dielectric layer **110** of the planar FET **102** and/or the gate dielectric layer **110** of the IO finFET **104io**. In some embodiments, the gate dielectric layer **110** of the core finFET **104c** is defined by the core dielectric layer **116** but not the first and second IO dielectric layers **112**, **114**. In alternative embodiments, the gate dielectric layer **110** of the core finFET **104c** is also defined by the first IO dielectric layer **112** and/or the second IO dielectric layer **114**.
- (31) The first IO dielectric layer **112** may, for example, be or comprise silicon oxide and/or some other suitable dielectric(s). Similarly, the second IO dielectric layer **114** may, for example, be or comprise silicon oxide and/or some other suitable dielectric(s). In some embodiments, the second IO dielectric layer **114** has an elevated concentration of nitrogen than the first IO dielectric layer **112**. The nitrogen may, for example, suppress diffusion of dopants through the first IO dielectric layer **112**, which may, for example, lead to reduced gate current leakage and/or increased reliability. In some embodiments, the first and second IO dielectric layers **112**, **114** are or comprise the same material, such that boundaries therebetween may not be readily distinguishable.
- (32) As seen hereafter, the first IO dielectric layer **112** may be formed without negatively impacting semiconductor devices on the semiconductor substrate **106**. For example, during the formation of the IC, dopants may be implanted into the semiconductor substrate **106** to tune threshold voltages and/or other suitable parameters of the semiconductor devices. The first IO dielectric layer **112** may be formed before this tuning so the deposition process employed for the first IO dielectric layer **112** does not impact these dopants. Because the first IO dielectric layer **112** may be formed without impacting the semiconductor devices, the first IO dielectric layer **112** and hence the planar FET **102** may be formed without negatively impacting yields for the semiconductor devices. The semiconductor devices may include, for example, SRAM devices, the core finFET **104c**, the IO finFET **104io**, some other suitable semiconductor device(s), or any combination of the foregoing. Further, the first IO dielectric layer **112** and hence the planar FET **102** may be integrated into existing finFET processes with no or minimal impact.
- (33) The core dielectric layer **116** may, for example, be or comprise silicon oxide, a high k dielectric, some other suitable dielectric(s), or any combination of the foregoing. In some embodiments, the core dielectric layer **116** has U-shaped profiles individual to and respectively cupping undersides of the gate electrodes **108**. In alternative embodiments, the core dielectric layer **116** has some other suitable profiles.
- (34) Hard masks **118** respectively cover the gate electrodes **108** and/or sidewall spacers **120** are

respectively on sidewalls of the gate electrodes **108**. In alternative embodiments, the hard masks **118** and/or the sidewall spacers **120** are omitted. The hard masks **118** may be or comprise, for example, silicon nitride and/or some suitable dielectric(s). Similarly, the sidewall spacers **120** may be or comprise, for example, silicon nitride and/or some other suitable dielectric(s). In some embodiments, the hard masks **118** and the sidewall spacers **120** are or comprise the same material, such that boundaries therebetween may not be readily distinguishable.

(35) Device wells **122** are respectively in the mesa **106m** and the fins **106f** and are respectively along top surfaces of the mesa **106m** and the fins **106f**. The device wells **122** are individual to the planar FET **102** and the finFETs **104** and respectively underlie the planar FET **102** and the finFETs **104**. The device wells **122** each have a doping type that varies depending upon a type (e.g., P-type or N-type) of a corresponding FET. For example, a P-type FET may have an N-type device well, whereas an N-type FET may have a P-type device well. In alternative embodiments, one, some, or all of the device wells **122** is/are omitted.

(36) Source/drain regions **124** respectively overlie the device wells **122** and are respectively in the mesa **106m** and the fins **106f**. Further, the source/drain regions **124** are arranged so that each of the gate electrodes **108** is sandwiched between two of the source/drain regions **124**. In embodiments, the source/drain regions **124** have individual source/drain extensions **124e**. The source/drain extensions **124e** respectively underlie the sidewall spacers **120** and have a lower doping concentration than remainders of corresponding source/drain regions. In alternative embodiments, some or all of the source/drain extensions **124e** are omitted. The source/drain regions **124** have opposite doping type as corresponding device wells.

(37) Channel regions **126** individual to the planar FET **102** and the finFETs **104** respectively underlie the corresponding FETs and extend between the source/drain regions of the corresponding FETs. Depending upon a bias voltage applied to the gate electrode of a FET, a corresponding channel region switches between a conducting state and a non-conducting state. In the conducting state, the source/drain regions of the FET are electrically coupled together. In the non-conducting state, the source/drain regions of the FET are electrically isolated.

(38) Epitaxial structures **128** respectively overlap with the source/drain regions **124** and impose stress on the channel regions **126** to increase carrier mobility. Source/drain regions that overlap with epitaxial structures may, for example, also be known as strained source/drain (SSD) regions. For an N-type FET, an epitaxial structure may, for example, be or comprise silicon carbide or some other suitable material that imposes tensile stress on a corresponding channel. For a P-type FET, an epitaxial structure may, for example, be or comprise silicon germanium or some other suitable material that imposes compressive stress on a corresponding channel. In some embodiments, the epitaxial structures **128** have hexagonal profiles, but other suitable profiles are amenable. In alternative embodiments, one, some, or all of the epitaxial structures **128** are omitted.

(39) As best seen in FIGS. **1C** and **1D**, the epitaxial structures **128** of the planar FET **102** have different profiles and heights than the epitaxial structures **128** of the finFETs **104**. For example, the epitaxial structures **128** of the planar FET **102** may have a first height $H_{sub.p}$, whereas the epitaxial structures **128** of the finFETs **104** may have a second height $H_{sub.fin}$ greater than the first height $H_{sub.p}$. The different profiles and heights may, for example, be due to different epitaxial loading during formation of the epitaxial structures **128**. For example, the epitaxial structures **128** of the planar FET **102** may be concurrently formed with the epitaxial structures **128** of the finFETs **104**. Different feature densities and/or pitches around locations of the epitaxial structures **128** may lead to different etch rates while forming openings within which the epitaxial structures **128** are formed. As a result, the openings may have different heights and profiles. In some embodiments, the epitaxial structures **128** of the planar FET **102** also have different volumes than the epitaxial structures **128** of the finFETs **104** due to, for example, the epitaxial loading.

(40) A trench isolation structure **130** overlies the semiconductor substrate **106** and surrounds the mesa **106m** and the fins **106f**. The trench isolation structure **130** steps down from the planar FET

102 to the finFETs **104**, such that the fins **106f** extend above the trench isolation structure **130**. Further, the trench isolation structure **130** has a recess **130r** that extends along a periphery of the mesa **106m** so a top surface of the mesa **106m** is even with or about even with a recessed top surface portion of the trench isolation structure **130**. In some embodiments, the recess **130r** extends in a closed path along the periphery of the mesa **106m**. In alternative embodiments, the recess **130r** is localized to edges of the mesa **106m** extending between the source/drain regions **124** of the planar FET **102**. The trench isolation structure **130** may be or comprise, for example, silicon oxide and/or some other suitable dielectric(s). Further, the trench isolation structure **130** may, for example, be a shallow trench isolation (STI) structure or some other suitable type of the trench isolation structure.

(41) It has been appreciated that the recess **130r** may, for example, enhance performance of the planar FET **102**. For example, edges of the mesa **106m** that extend between the source/drain regions **124** of the mesa **106m** may have a high concentration of defects that negatively affect the saturation current and off current for the planar FET **102**. The recess **130r** changes the structure at the edges to at least partially mitigate the effect of the defects and to enhance the saturation current and off current for the planar FET **102**. This may, in turn, enhance the reliability of the planar FET **102**. In some embodiments, a distance D.sub.1 from the top surface of the mesa **106m** to the top surface of the trench isolation structure **130** is greater than about 4 nanometers, 6 nanometers, or some other suitable value. In alternative embodiments, the distance D.sub.1 is about 2-4 nanometers, about 2-6 nanometers, less than or equal to about 4 nanometers, less than or equal to about 6 nanometers, or some other suitable value. In at least some embodiments in which the distance D.sub.1 is less than or equal to about 4 nanometers, 6 nanometers, or some other suitable value, the recess **130r** may be omitted because of a minimal performance improvement.

(42) While the planar FET **102** is described as being employed for IO, it is to be appreciated that the planar FET **102** is not limited to IO. Rather, the planar FET **102** may alternatively be used for logic, HV applications, and so on in alternative embodiments. Similarly, while the core finFET **104c** and the IO finFET **104io** are respectively described with regard to a logic core and IO, the core finFET **104c** and the IO finFET **104io** are not limited to a logic core and IO. The core finFET **104c** and/or the IO finFET **104io** may be used for other suitable applications in alternative embodiments. For example, the core finFET **104c** may be used for SRAM. Further yet, while the planar FET **102** is being integrated with finFETs, the planar FET **102** may alternatively be integrated with nanosheet FETs, gate-all-around (GAA) FETs, and so on. For example, the core finFET **104c** may be replaced with a nanosheet FET, a GAA FET, or another suitable type of FET, and/or the IO finFET **104io** may be replaced with a nanosheet FET, a GAA FET, or another suitable type of FET.

(43) With reference to FIG. 2, a perspective view **200** of some embodiments of the IC of FIGS. 1A-1D is provided. For clarity, the fins **106f**, the mesa **106m**, and the source/drain regions **124** are shown in phantom where covered by the trench isolation structure **130**.

(44) With reference to FIGS. 3A-3E, cross-sectional views **300A-300E** of some alternative embodiments of the IC of FIGS. 1A-1D is provided. The cross-sectional views **300A-300E** may, for example, be taken along line A-A' in FIG. 1A.

(45) As illustrated by the cross-sectional view **300A** of FIG. 3A, deep wells **302** are respectively in the mesa **106m** and the fins **106f**. Further, the deep wells **302** respectively underlie the device wells **122**. Although not visible, one of the deep wells **302** may underlie the device well **122** of the core finFET **104c** as illustrated for the IO finFET **104io**. The deep wells **302** have opposite doping types as corresponding device wells. The deep wells **302** may, for example, suppress diffusion of dopants and/or impurities in the device wells **122** to a remainder of the semiconductor substrate **106** or vice versa. Additionally, or alternatively, the deep wells **302** may, for example, provide enhanced electrical isolation to the planar FETs **102** and the finFETs **104**. In some embodiments, the deep wells **302** are employed for N-type FETs, but not P-type FETs. In alternative embodiments, one,

some, or all of the deep wells **302** is/are omitted.

(46) As illustrated by the cross-sectional view **300B** of FIG. **3B**, a top surface of the mesa **106m** is recessed relative to top surfaces of the fins **106f** by a distance $D_{sub.2}$. In alternative embodiments, the top surface of the mesa **106m** may be elevated relative to the top surfaces of the fins **106f** the distance $D_{sub.2}$. The vertical offset between the top surface of the mesa **106m** and the top surfaces of the fins **106f** may, for example, be employed to minimize uneven loading of a planarization process performed while forming the IC. Such uneven loading could lead to an uneven planarization, which could lead to misalignment during photolithography and other suitable semiconductor manufacturing processes performed while forming the IC.

(47) As illustrated by the cross-sectional view **300C** of FIG. **3C**, the recess **130r** (see, e.g., FIG. **1B**) at a periphery of the mesa **106m** is omitted. As such, a sidewall of the trench isolation structure **130** that adjoins the mesa **106m** has a top edge elevated above a top surface of the mesa **106m**. The recess **130r** may, for example, enhance performance of the planar FET **102**. However, where the distance $D_{sub.1}$ from the top surface of the mesa **106m** to the top surface of the trench isolation structure **130** is less than about 4 nanometers, 6 nanometers, or some other suitable value, the performance improvement from the recess **130r** may be minimal and hence the recess **130r** may be omitted with minimal impact on performance of the planar FET **102**. As seen hereafter, this may lead to a cost savings. In some embodiments, the distance $D_{sub.1}$ is about 2-4 nanometers, about 2-6 nanometers, about 4-6 nanometers, or some other suitable value.

(48) As illustrated by the cross-sectional view **300D** of FIG. **3D**, the recess **130r** (see, e.g., FIG. **1B**) at a periphery of the mesa **106m** is omitted and the distance $D_{sub.1}$ (see, e.g., FIG. **1B**) is zero or about zero. As such, the gate electrode **108** of the planar FET **102** has a bottom profile that is flat and/or planar. Other suitable profiles are, however, amenable. Additionally, the trench isolation structure **130** steps down from the planar FET **102** to the finFETs **104** at a location closer to the planar FET **102** and aligned to a sidewall spacer of the planar FET **102**. In alternative embodiments, the trench isolation structure **130** steps down from the planar FET **102** to the finFETs **104** at the location illustrated in FIG. **1B** or at some other suitable location.

(49) As illustrated by the cross-sectional view **300E** of FIG. **3E**, the core dielectric layer **116** may, for example, be omitted from the gate dielectric layer **110** of the planar FET **102**.

(50) While FIGS. **3A-3E** describe variations to the IC of FIGS. **1A-1D** through modification of the cross-sectional view **100B** of FIG. **1B**, it is to be appreciated that the top and cross-sectional views **100A**, **100C**, and **100D** respectively at FIGS. **1A**, **1C**, and **1D** may be similarly modified. For example, the cross-sectional view **100D** of FIG. **1D** may be modified to include deep wells **302** respectively underlying the core finFET **104c** and the IO finFET **104io** as illustrated in FIG. **3A**. Further, while FIGS. **3A-3E** describe different variations to the IC of FIGS. **1A-1D**, it is to be appreciated that combinations of the variations are amenable. For example, the IC may have deep wells **302** as illustrated in FIG. **3A** and may further omit the recess **130r** (see, e.g., FIG. **1B**) at a periphery of the mesa **106m** as illustrated in FIG. **3C**.

(51) With reference to FIGS. **4A** and **4B**, cross-sectional views **400A**, **400B** of some alternative embodiments of the IC of FIGS. **1A-1D** are provided. The cross-sectional views **400A** and **400B** may, for example, be taken along line B-B' in FIG. **1A**.

(52) As illustrated by the cross-sectional view **400A** of FIG. **4A**, the epitaxial structures **128** comprise corresponding first epitaxial layers **128a** and corresponding second epitaxial layers **128b**. In alternative embodiments, the epitaxial structures **128** comprise more or less epitaxial layers. The first epitaxial layers **128a** define a periphery of the epitaxial structures **128** and separate the second epitaxial layers **128b** from the semiconductor substrate **106**. Further, the first and second epitaxial layers **128a**, **128b** have different material compositions. By forming the epitaxial structures **128** with multiple epitaxial layers with different material compositions, material compositions of the epitaxial structures **128** may be finely tuned.

(53) As illustrated by the cross-sectional view **400B** of FIG. **4B**, multiple planar FETs **102** are on

the mesa **106m** (e.g., instead of a single planar FET). While two planar FETs **102** are illustrated, more planar FETs are amenable. The planar FETs **102** are each as the planar FET **102** of FIG. **1C** is illustrated and described, except that the planar FETs **102** share a common one of the device wells **122** and a common one of the source/drain regions **124**.

(54) While FIGS. **4A** and **4B** describe variations to the IC of FIGS. **1A-1D** through modification of the cross-sectional view **100C** of FIG. **1C**, it is to be appreciated that the top and cross-sectional views **100A**, **100B**, and **100D** respectively at FIGS. **1A**, **1B**, and **1D** may be similarly modified. Further, while FIGS. **4A** and **4B** describe different variations to the IC of FIGS. **1A-1D**, it is to be appreciated that combinations of the variations are amenable.

(55) With reference to FIG. **5**, a cross-sectional view **500** of some embodiments of an IC comprising a planar FET **102** and an IO finFET **104io** that are spaced from each other in a direction extending along corresponding channel lengths is provided. The planar FET **102** and the IO finFET **104io** are as described and illustrated at FIGS. **1A-1D**, except that the planar FET **102** and the IO finFET **104io** have different relative positioning.

(56) The epitaxial structures **128** of the planar FET **102** have different profiles and heights than the epitaxial structures **128** of the IO finFET **104io**. For example, the epitaxial structures **128** of the planar FET **102** may have a first height $H_{sub.p}$, whereas the epitaxial structures **128** of the IO finFET **104io** may have a second height $H_{sub.fin}$ greater than the first height $H_{sub.p}$. The different profiles and heights may, for example, be due to different epitaxial loading during formation of the epitaxial structures **128**. For example, the epitaxial structures **128** of the planar FET **102** may be concurrently formed with the epitaxial structures **128** of the IO finFET **104io**. Because the mesa **106m** may be larger than the fin **106f**, the feature densities and exposed surface areas may be different at the mesa **106m** and the fin **106f**. The different feature densities and exposed surface areas may lead to different etch rates while forming openings within which the epitaxial structures **128** are formed. As a result of the different etch rates, the openings within which the epitaxial structures **128** of the planar FET **102** are formed may be shallower than the openings within which the epitaxial structures **128** of the IO finFET **104io** are formed.

(57) With reference to FIGS. **6A** and **6B**, cross-sectional views **600A**, **600B** of some alternative embodiments of the IC of FIG. **5** is provided.

(58) As illustrated by the cross-sectional view **600A** of FIG. **6A**, the epitaxial structures **128** comprise corresponding first epitaxial layers **128a** and corresponding second epitaxial layers **128b**. In alternative embodiments, the epitaxial structures **128** comprise more epitaxial layers. The first epitaxial layers **128a** define a periphery of the epitaxial structures **128** and separate the second epitaxial layers **128b** from the semiconductor substrate **106**. Further, the first and second epitaxial layers **128a**, **128b** have different material compositions. As discussed previously, by forming the epitaxial structures **128** with multiple epitaxial layers with different material compositions, material compositions of the epitaxial structures **128** may be finely tuned.

(59) The first and second epitaxial layers **128a**, **128b** of the planar FET **102** have different profiles and thicknesses than the IO finFET **104io**. For example, the first epitaxial layer **128a** of the planar FET **102** may have a first thickness $T_{sub.p}$ at sidewalls, whereas the first epitaxial layer **128a** of the IO finFET **104io** may have a second thickness $T_{sub.fin}$ greater than the first thickness $T_{sub.p}$ at sidewalls. The different profiles and thicknesses may, for example, be due to different epitaxial loading during formation of the epitaxial structures **128**. For example, different feature densities and/or pitches around locations of the epitaxial structures **128** may lead to different etch rates while forming openings within which the epitaxial structures **128** are formed. As a result, the openings may have different heights and profiles. The different heights and profiles of the openings may lead to different epitaxial deposition rates while depositing the first and second epitaxial layers **128a**, **128b**. For example, surfaces with different orientations may have different deposition rates. As a result, the first and second epitaxial layers **128a**, **128b** of the planar FET **102** may be deposited with different profiles and thicknesses than the IO finFET **104io**. In some embodiments, the first

and second epitaxial layers **128a**, **128b** of the planar FET **102** may be deposited with different volumes than the IO finFET **104io**.

(60) Because the first and second epitaxial layers **128a**, **128b** of the planar FET **102** have different profiles and thickness than the IO finFET **104io**, the epitaxial structures **128** of the planar FET **102** have different material compositions than the epitaxial structures **128** of the IO finFET **104io**. For example, a ratio between material of the first epitaxial layers **128a** to material of the second epitaxial layers **128b** may be different at the epitaxial structures **128** of the planar FET **102** than at the epitaxial structures **128** of the IO finFET **104io**.

(61) As illustrated by the cross-sectional view **600B** of FIG. **6B**, the gate length $L_{sub.g}$ of the planar FET **102** is reduced so it's the same as or substantially the same as that of the IO finFET **104io**. Further, the layout of the trench isolation structure **130** is varied.

(62) With reference to FIG. **7A**, a cross-sectional view **700A** of some more detailed embodiments of the IC of FIG. **5** is provided in which the IC further comprises a memory array **702** and an interconnect structure **704** electrically coupled to the planar FET **102**, the IO finFET **104io**, and the memory array **702**. The memory array **702** comprises a plurality of memory cells **706**. The memory cells **706** are in a plurality of columns and, although not visible in the cross-sectional view **700A**, may be in a plurality of rows. The memory cells **706** may, for example, be resistive random-access memory (RRAM) cells, magnetoresistive random-access memory (MRAM) cells, or some other suitable type of memory cells.

(63) The interconnect structure **704** comprises an interconnect dielectric layer **708**, and further comprises a plurality of contact vias **710**, a plurality of wires **712**, and a plurality of interwire vias **714**. The interconnect dielectric layer **708** may be or comprise, for example, silicon oxide, a low k dielectric, some other suitable dielectric(s), or any combination of the foregoing. The contact vias **710**, the wires **712**, and the interwire vias **714** are alternately stacked in the interconnect dielectric layer **708** to define conductive paths leading from the planar FETs **102** and the IO finFET **104io**. Such a conductive path may extend from a source/drain region of the planar FET **102** to a bit line BL of the memory array **702**, which may be defined by one of the wires **712**. The contact vias **710**, the wires **712**, and the interwire vias **714** may, for example, be or comprise metal and/or some other suitable conductive material(s).

(64) A contact etch stop layer (CESL) **716** is between the trench isolation structure **130** and the interconnect structure **704** to provide an etch stop while forming the contact vias **710**. For example, the CESL **716** may provide an etch stop while forming contact vias extending to the source/drain regions **124** of the planar FET **102** and the IO finFET **104io**. The CESL **716** may, for example, be or comprise silicon nitride and/or some other suitable dielectric(s).

(65) With reference to FIG. **7B**, a cross-sectional view **700B** of some alternative embodiments of the IC of FIG. **7A** is provided in which the memory array **702** is omitted. Further, a plurality of pads **718** and a passivation layer **720** are along a top of the interconnect dielectric layer **708**. The pads **718** may, for example, be employed for IO, ground, power, or some other application. The pads **718** are electrically coupled to the planar FET **102** and the IO finFET **104io** by the contact vias **710**, the wires **712**, and the interwire vias **714**. The passivation layer **720** overlies the interconnect dielectric layer **708** and defines pad openings **722** exposing the pads **718**. The pad openings **722** may, for example, accommodate solder bumps or some other suitable conductive structures electrically coupling the pads **718** to structure external to the IC. The passivation layer **720** may, for example, be or comprise silicon oxide, silicon nitride, some other suitable dielectric(s), or any combination of the foregoing.

(66) While FIGS. **5**, **6A**, **6B**, **7A**, and **7B** are described with the IO finFET **104io**, a core finFET **104c** may replace the IO finFET **104io** in alternative embodiments. The core finFET **104c** may, for example, be as described and illustrated at FIGS. **1A-1D**. Further, the planar FET **102** and/or the IO finFET **104io** in any one of FIGS. **5**, **6A**, **6B**, **7A**, and **7B** may be modified as described and illustrated in any one or combination FIGS. **3A-3E**, **4A**, and **4B** in alternative embodiments.

Further, the memory array **702** of FIG. 7A may be omitted in alternative embodiments.

(67) With reference to FIGS. **8A** and **8B**, top and cross-sectional views **800A**, **800B** of some embodiments of an IC comprising a planar FET **102** and a plurality of finFETs **104** spaced in a direction extending transverse to corresponding channel lengths is provided. The cross-sectional view **800B** of FIG. **8B** may, for example, be taken along line D-D' in FIG. **8A**.

(68) The planar FET **102** and the finFETs **104** are as described at FIGS. **1A-1D** except that the finFETs **104** share a gate stack **802** and certain features (e.g., device wells, the trench isolation structure, etc.) are omitted. These omitted features may be present in alternative embodiments. The gate stack **802** comprises a gate dielectric layer (see, e.g., **110** at FIGS. **1A-1D**), a gate electrode (see, e.g., **108** at FIGS. **1A-1D**) overlying the gate dielectric layer, and a hard mask **118** overlying the gate electrode. In alternative embodiments, at least one of the finFETs **104** has an individual gate stack and/or at least two of the finFETs **104**, but less than all of the finFETs **104**, share a gate stack **802** with each other but not a remainder of the finFETs **104**.

(69) The epitaxial structures **128** of the planar FET **102** (e.g., planar epitaxial structures) have a first height $H_{\text{sub.p}}$, whereas the epitaxial structures **128** of the finFETs **104** (e.g., fin epitaxial structures) have a second height $H_{\text{sub.fin}}$ greater than the first height $H_{\text{sub.p}}$. Further, top surfaces of the planar epitaxial structures are recessed relative to top surfaces of the fin epitaxial structures, and bottom surfaces of the planar epitaxial structures are elevated above bottom surfaces of the fin epitaxial structures. As described above, this variation between the epitaxial structures **128** may be due to different epitaxial loads at the planar FET **102** and at the finFETs **104**.

(70) With reference to FIGS. **9A** and **9B**, top and cross-sectional views **900A**, **900B** of some embodiments of an IC comprising a plurality of planar FETs **102** and a plurality of finFETs **104** that are electrically coupled to an interconnect structure **704** is provided. The cross-sectional view **900B** of FIG. **9B** may, for example, be taken along line E-E' in FIG. **9A**, whereas the top view **900A** of FIG. **9A** may, for example, be taken along line F-F' in FIG. **9B**.

(71) The plurality of planar FETs **102** comprises an N-type planar FET **102_n** and a P-type planar FET **102_p**. Further, the plurality of finFETs **104** comprises an N-type IO finFET **104_{ion}**, an N-type core finFET **104_{cn}**, a P-type IO finFET **104_{iop}**, and a P-type core finFET **104_{cp}**. The N-type planar FET **102_n**, the N-type IO finFET **104_{ion}**, and the N-type core finFET **104_{cn}** are respectively as the planar FET **102**, the IO finFET **104_{io}**, and the core finFET **104_c** are illustrated and described at FIGS. **1A-1D**, except that these N-type FETs respectively have N-type gate electrodes **108_n** and N-type source/drain regions **124_n** (shown in phantom). Further, these N-type FETs have P-type device wells **122_p** and deep N wells **302_n** respectively underlying the P-type device wells **122_p**. The P-type planar FET **102_p**, the P-type IO finFET **104_{iop}**, and the P-type core finFET **104_{cp}** are respectively as the planar FET **102**, the IO finFET **104_{io}**, and the core finFET **104_c** are illustrated and described at FIGS. **1A-1D**, except that these P-type FETs respectively have P-type gate electrodes **108_p** and P-type source/drain regions **124_p** (shown in phantom). Further, these P-type FETs have N-type device wells **122_n**.

(72) The N-type gate electrodes **108_n** are gate electrodes having an N-type work function. An n-type work function may, for example, be: 1) a work function within about 0.1 electron volts (eV), about 0.2 eV, or about 0.4 eV of a work function for N-type polycrystalline or monocrystalline silicon; 2) a work function less than about 4.0, about 4.2, or about 4.4 eV; 3) a work function between about 3.5-4.4 eV, about 4.0-4.4 eV, or about 3.8-4.5 eV; 4) some other suitable n-type work function; or 5) any combination of the foregoing. The P-type gate electrodes **108_p** are gate electrodes having a P-type work function. A P-type work function may, for example, be: 1) a work function within about 0.1, 0.2, or 0.4 eV of a work function for p-type polycrystalline or monocrystalline silicon; 2) a work function greater than about 4.8 eV, about 5.0 eV, or about 5.2 eV; 3) a work function between about 4.8-5.2 eV, about 5.0-5.4 eV, or about 4.6-5.6 eV; 4) some other suitable p-type work function; or 5) any combination of the foregoing. The N-type and P-type gate electrode **108_p** may, for example, be or comprise metal, doped polysilicon, some other suitable

conductive materials, or any combination of the foregoing.

(73) The interconnect structure **704** comprises an interconnect dielectric layer **708**, and further comprises a plurality of contact vias **710**, a plurality of wires **712**, and a plurality of interwire vias **714** stacked in the interconnect dielectric layer **708**. Further, the interconnect structure **704** is separated from the trench isolation structure **130** by a CESL **716**. The interconnect structure **704** and the CESL **716** may, for example, be as described with regard to FIG. 7A. In some embodiments, the interconnect structure **704** further comprises a passivation layer **720**, pads **718**, and corresponding pad openings **722** in the passivation layer **720** as illustrated at FIG. 7B. In some embodiments, the pads **718** are electrically coupled to the planar FETs **102** and/or the finFETs **104** by the contact vias **710**, the wires **712**, and the interwire vias **714**.

(74) With reference to FIGS. **10A** and **10B** through FIGS. **13A** and **13B**, FIGS. **14A-14C** through FIGS. **23A-23C**, FIGS. **24A** and **24B**, FIGS. **25A-25D** through FIGS. **28A-28D**, FIGS. **29A** and **29B** through FIGS. **37A** and **37B**, a series of views of some embodiments of a method for forming an IC comprising a planar FET and a finFET is provided. Figures labeled with a suffix of B illustrate cross-sectional views along line E-E' in like-numbered figures with a suffix of A. For FIGS. **14A-14C** through FIGS. **23A-23C**, figures labeled with a suffix of C illustrate perspective views of like-numbered figures with suffixes of A and B. For FIGS. **25A-25D** through FIGS. **28A-28D**, figures labeled with a suffix of C illustrate cross-sectional views along line G-G' or H-H' (whichever is present) in like-numbered figures with a suffix of A. For FIGS. **25A-25D** through FIGS. **28A-28D**, figures labeled with a suffix of D illustrate cross-sectional views along line I-I', J-J', or K-K' (whichever is present) in like-numbered figures with a suffix of A. The method is illustrated by formation of the IC of FIGS. **9A** and **9B** but may also be employed to form the IC in any one or combination of FIGS. **1A-1D**, **2**, **3A-3E**, **4A**, **4B**, **5**, **6A**, **6B**, **7A**, **7B**, **8A**, and **8B**. Further, the method may be employed to form other suitable ICs.

(75) As illustrated by the views **1000A**, **1000B** of FIGS. **10A** and **10B**, a first pad layer **1002**, a second pad layer **1004**, and a first hard mask layer **1006** are deposited over a semiconductor substrate **106**. The first pad layer **1002** and/or the first hard mask layer **1006** may, for example, be or comprise silicon oxide and/or some other suitable dielectric(s). The second pad layer **1004** may, for example, be or comprise silicon nitride and/or some other suitable dielectric(s). In some embodiments, the first pad layer **1002** and the first hard mask layer **1006** are or comprise silicon oxide, and the second pad layer **1004** is or comprise silicon nitride.

(76) Also illustrated by the views **1000A**, **1000B** of FIGS. **10A** and **10B**, a second hard mask layer **1008** is deposited over the first hard mask layer **1006** and patterned with a pattern for fins hereafter formed. The second hard mask layer **1008** may, for example, be or comprise silicon nitride and/or some other suitable dielectric(s). In some embodiments, the second hard mask layer **1008** is or comprise the same material as the second pad layer **1004**.

(77) As illustrated by the views **1100A**, **1100B** of FIGS. **11A** and **11B**, the first hard mask layer **1006** is patterned with a pattern for fins and mesas hereafter formed. Such patterning may, for example, comprise: 1) forming a photoresist mask **1102** having a pattern for the mesas using photolithography; 2) performing an etch into the first hard mask layer **1006** using the photoresist mask **1102** and the second hard mask layer **1008** as a mask; and 3) removing the photoresist mask **1102**. Other suitable patterning processes are, however, amenable.

(78) As illustrated by the views **1200A**, **1200B** of FIGS. **12A** and **12B**, a first etch is performed into the semiconductor substrate **106** and the first and second pad layers **1002**, **1004** with the first and second hard mask layers **1006**, **1008** (see, e.g., FIGS. **11A** and **11B**) in place to form a plurality of mesas **106m** and a plurality of fins **106f**. As seen hereafter, the mesas **106m** correspond to an N-type planar FET and a P-type planar FET being formed. Further, the fins **106f** correspond to an N-type IO finFET, an N-type core finFET, a P-type IO finFET, and a P-type core finFET being formed. In alternative embodiments, the mesas **106m** and/or the fins **106f** correspond to different device types and/or combinations of device types. For example, the mesas **106m** may correspond to

N-type planar FETs, but not P-type planar FETs, in alternative embodiments.

(79) Also illustrated by the views **1200A**, **1200B** of FIGS. **12A** and **12B**, the first and second hard mask layers **1006**, **1008** (see, e.g., FIGS. **11A** and **11B**) are removed. The removal may, for example, be performed as part of the first etch. Alternatively, the removal may, for example, be performed independent of and after the first etch.

(80) As illustrated by the views **1300A**, **1300B** of FIGS. **13A** and **13B**, a trench isolation structure **130** is formed around the mesas **106m** and the fins **106f**. A process for forming the trench isolation structure **130** may, for example, comprise: 1) depositing a dielectric layer covering and surrounding the mesas **106m** and the fins; and 2) performing a planarization into the dielectric layer until a top surface of the dielectric layer is about even with a top surface of the second pad layer **1004**. Other suitable processes are, however, amenable.

(81) As illustrated by the views **1400A-1400C** of FIGS. **14A-14C**, a second etch is performed into the trench isolation structure **130** to recess a top surface of the trench isolation structure **130** relative to a top surface of the second pad layer **1004** by a distance $D_{sub.3}$. The recessing may, for example, enhance subsequent removal of second pad layer **1004** by exposing sidewalls of the second pad layer **1004** to etchants used for the removal. The second etch may, for example, be performed by a non-plasma dry etch and/or some other suitable etch type. In some embodiments, but not all embodiments, the second etch is anisotropic.

(82) As illustrated by the views **1500A-1500C** of FIGS. **15A-15C**, a third etch is performed into the second pad layer **1004** (see, e.g., FIGS. **14A-14C**) to remove the second pad layer **1004** and to expose the first pad layer **1002**. The third etch may, for example, be performed by a wet etch or some other suitable type of etch.

(83) As illustrated by the views **1600A-1600C** of FIGS. **16A-16C**, deep N wells **302n** are formed respectively in the fins **106f** for the N-type finFETs being formed and the mesa **106m** for the N-type planar FET being formed. In alternative embodiments, the deep N wells **302n** are not formed. A process for forming the deep N wells **302n** may, for example, comprise: 1) forming a photoresist mask **1602** masking the fins **106f** for the P-type finFETs being formed and the mesa **106m** for the P-type planar FET being formed using photolithography; 2) implanting dopants into the semiconductor substrate **106** with the photoresist mask **1602** in place; 3) removing the photoresist mask **1602**; and 4) activating the dopants by annealing. Other suitable processes are, however, amenable. In some embodiments, the process is preceded by a photoresist cleaning process. The photoresist cleaning process and/or the removing of the photoresist mask **1602** may, for example, be performed by a Caro's acid strip or some other suitable process.

(84) As illustrated by the views **1700A-1700C** of FIGS. **17A-17C**, a P-type device well **122p** is formed overlying a corresponding deep N well **302n** in the mesa **106m** for the N-type planar FET being formed. A process for forming the P-type device well **122p** may, for example, comprise: 1) forming a photoresist mask **1702** masking the fins **106f** and the mesa **106m** for the P-type planar FET being formed using photolithography; 2) implanting dopants into the semiconductor substrate **106** with the photoresist mask **1702** in place; and 3) removing the photoresist mask **1702**. Other suitable processes are, however, amenable. The photoresist mask **1702** may, for example, be removed using a Caro's acid strip or some other suitable process.

(85) As illustrated by the views **1800A-1800C** of FIGS. **18A-18C**, an N-type device well **122n** is formed in the mesa **106m** for the P-type planar FET. A process for forming the N-type device well **122n** may, for example, comprise: 1) forming a photoresist mask **1802** masking the fins **106f** and the mesa **106m** for the N-type planar FET being formed using photolithography; 2) implanting dopants into the semiconductor substrate **106** with the photoresist mask **1802** in place; and 3) removing the photoresist mask **1802**. Other suitable processes are, however, amenable. The photoresist mask **1802** may, for example, be removed using a Caro's acid strip or some other suitable process.

(86) As illustrated by the views **1900A-1900C** of FIGS. **19A-19C**, a fourth etch is performed into

the first pad layer **1002** (see, e.g., FIGS. **18A-18C**) to remove the first pad layer **1002**. The fourth etch may, for example, be performed by a dry etch or some other suitable type of etch.

(87) As illustrated by the views **2000A-2000C** of FIGS. **20A-20C**, the trench isolation structure **130** is patterned to form recesses **130r** that extend in closed paths respectively along peripheries of the mesa **106m**. The recesses **130r** are formed so top surfaces of the mesas **106m** are even with or about even with recessed top surface portions of the trench isolation structure **130** and sidewalls of the trench isolation structure **130** that adjoin the mesas **106m** have top edges even with or about even with the top surfaces of the mesas **106m**. It has been appreciated this may, for example, enhance performance of the planar FETs being formed on the mesas **106m**. For example, edges of the mesas **106m** extending along channel lengths of the planar FETs may have a high concentration of defects that negatively affect the saturation current and off current for the planar FETs. The recesses **130r** change the structure at the edges to at least partially mitigate the effect of the defects and to enhance the saturation current and off current for the planar FETs. This may, in turn, enhance reliability of the planar FETs.

(88) In some embodiments, a distance D.sub.1 from the top surfaces of the mesas **106m** to the top surface of the trench isolation structure **130** is greater than about 4 nanometers, about 6 nanometers, or some other suitable value. In alternative embodiments, the distance D.sub.1 is less than or equal to about 4 nanometers, about 6 nanometers, or some other suitable value. In at least some of such embodiments, the patterning and the resulting recesses **130r** may be omitted because of a minimal performance improvement. This may lead to a cost savings.

(89) The patterning may, for example, comprise: 1) forming a photoresist mask **2002** having a pattern for the recesses **130r** using photolithography; 2) performing an etch into the trench isolation structure **130** with the photoresist mask **2002** in place; and 3) removing the photoresist mask **2002**. Other suitable patterning processes are, however, amenable. The etch may, for example, be performed by a wet etch or some other suitable type of etch. The photoresist mask **2002** may, for example, be removed using a Caro's acid strip or some other suitable process.

(90) As illustrated by the views **2100A-2100C** of FIGS. **21A-21C**, a first IO dielectric layer **112** is deposited over and lining the trench isolation structure **130** and the semiconductor substrate **106**. As seen hereafter, the first IO dielectric layer **112** may, for example, substantially define gate dielectric layers for planar FETs being formed on the mesas **106m**.

(91) In some embodiments, the first IO dielectric layer **112** is deposited by in situ steam generated (ISSG) oxidation or some other suitable thermal oxidation process. In alternative embodiments, the first IO dielectric layer **112** is deposited without thermal oxidation by a vapor deposition process or some other suitable process. In some embodiments, the deposition is preceded by a B clean or some other suitable cleaning process. In at least some embodiments in which the first IO dielectric layer **112** is deposited by a thermal oxidation process, a thickness T.sub.io1 of the first IO dielectric layer **112** is greater on the semiconductor substrate **106** than on the trench isolation structure **130**. In some embodiments, the thickness T.sub.io1 of the first IO dielectric layer **112** is about 50-70 angstroms, about 60 angstroms, or some other suitable value where the first IO dielectric layer **112** is on the semiconductor substrate **106**.

(92) As illustrated by the views **2200A-2200C** of FIGS. **22A-22C**, additional P-type device wells **122p** are formed overlying corresponding deep N wells **302n** in the fins **106f** for the N-type finFET being formed. Further, additional N-type device wells **122n** are formed in the fins **106f** for the P-type finFETs being formed. The additional P-type device wells **122p** may, for example, be formed as illustrated and described at FIGS. **17A-17C** using an appropriately patterned photoresist mask, whereas the additional N-type device wells **122n** may, for example, be formed as illustrated and described at FIGS. **18A-18C** using an appropriately patterned photoresist mask.

(93) Also illustrated by the views **2200A-2200C** of FIGS. **22A-22C**, the first IO dielectric layer **112** and the trench isolation structure **130** are patterned to clear the first IO dielectric layer **112** from the fins **106f** and to recess the trench isolation structure **130** around the fins. As a result, the fins **106f**

extend above a top surface of the trench isolation structure **130** by a distance D.sub.4. The patterning may, for example, comprise: 1) forming a photoresist mask **2202** covering the mesas **106m** and neighboring portions of the trench isolation structure **130** using photolithography; 2) performing an etch into the first IO dielectric layer **112** and the trench isolation structure **130** with the photoresist mask **2202** in place; and 3) removing the photoresist mask **2202**. Other suitable patterning processes are, however, amenable. The etch may, for example, be performed by a dry etch or some other suitable type of etch. The photoresist mask **2202** may, for example, be removed using a Caro's acid strip or some other suitable process. In some embodiments, a B clean or some other suitable cleaning process is performed after removing the photoresist mask **2202**.

(94) While not illustrated, dopants may be further implanted into semiconductor substrate to tune parameters of FETs being formed on the semiconductor substrate **106**. The tuned FETs may, for example, be or comprise the FETs being formed on the mesas **106m** and/or the fins **106f**. Further, the tuned FETs may, for example, be or comprise other finFETs and/or other planar FETs that are not shown. The other finFETs may, for example, define an SRAM array or some other suitable functional block of the IC.

(95) As seen hereafter, the first IO dielectric layer **112** substantially defines gate dielectric layers for planar FETs being formed. Further, in some embodiments, but not all embodiments, the first IO dielectric layer **112** is only used for the gate dielectric layers of the planar FETs. By forming the first dielectric layer before the recessing of the trench isolation structure **130** around the fins **106f**, the first IO dielectric layer **112** may be formed without negatively impacting doping profiles of the tuned FETs. For example, the first IO dielectric layer **112** may be formed by thermal oxidation for high quality and low leakage. Thermal oxidation may consume the semiconductor substrate **106** at the fins **106f** and the mesas **106m**, as well as at other locations on the semiconductor substrate **106**. If the first IO dielectric layer **112** was formed after the recessing and after the tuning, thermal oxidation may consume portions of the semiconductor substrate **106** at which dopants are implanted for the tuning. This, in turn, may change doping profiles of the tuned FETs and may hence change the threshold voltages and/or other suitable parameters. Because the first IO dielectric layer **112** may be formed without negatively impacting doping profiles, the first IO dielectric layer **112** and hence the planar FETs being formed may be formed without negatively impacting yields for the tuned FETs. Further, the planar FETs may be integrated with existing finFET process flows without impacting yields.

(96) As illustrated by the views **2300A-2300C** of FIGS. **23A-23C**, a second IO dielectric layer **114** is deposited lining the trench isolation structure **130** and the semiconductor substrate **106** over the first IO dielectric layer **112**. As seen hereafter, the second IO dielectric layer **114** may, for example, substantially or wholly define gate dielectric layers for IO finFETs being formed. Further, the second IO dielectric layer **114** may, for example, partially define gate dielectric layers for planar FETs being formed on the mesas **106m**.

(97) In some embodiments, the second IO dielectric layer **114** is deposited by wet oxidation or some other suitable thermal oxidation process. In alternative embodiments, the second IO dielectric layer **114** is deposited without thermal oxidation by a vapor deposition process or some other suitable process. In at least some embodiments in which the second IO dielectric layer **114** is deposited by a thermal oxidation process, a thickness T.sub.io2 of the second IO dielectric layer **114** is greater on the semiconductor substrate **106** than on the trench isolation structure **130** and on the first IO dielectric layer **112**. In some embodiments, the thickness T.sub.io2 of the second IO dielectric layer **114** is about 30-50 angstroms, about 38 angstroms, or some other suitable value where the second IO dielectric layer **114** is on the semiconductor substrate **106**. In some embodiments, the thickness T.sub.io2 of the second IO dielectric layer **114** is about 1-10 angstroms, about 3 angstroms, or some other suitable value where the second IO dielectric layer **114** is on the trench isolation structure **130** and/or on the first IO dielectric layer **112**. In some embodiments, a combined thickness T.sub.io of the first and second IO dielectric layers **112**, **114** is about 60-70

angstroms, about 63 angstroms, or some other suitable value at the mesas **106m**.

(98) In some embodiments, the deposition is preceded by a B clean or some other suitable cleaning process. In some embodiments, the deposition is proceeded by decoupled plasma nitridation (DPN) and a post nitridation annealing (PNA). The nitridation and corresponding annealing may, for example, suppress dopant diffusion through the first and/or second IO dielectric layers **114** to reduce gate leakage current and/or increase reliability of FETs being formed. In some embodiments, because of the DPN and the PNA, the second IO dielectric layer **114** has a higher concentration of nitrogen than the first IO dielectric layer **112**.

(99) As illustrated by the views **2400A**, **2400B** of FIGS. **24A** and **24B**, polysilicon gate electrodes **108ps** and corresponding hard masks **2402** are formed respectively on the mesas **106m** and the fins **106f**. In alternative embodiments, gate electrodes being or comprising a material other than polysilicon are used in place of the polysilicon gate electrodes **108ps**. A process for forming the polysilicon gate electrodes **108ps** and the hard masks **2402** may, for example, comprise: 1) depositing a polysilicon layer covering the first and second IO dielectric layers **112**, **114**; 2) depositing a hard mask layer covering the polysilicon layer; and 3) patterning the polysilicon layer and the hard mask layer by a photolithography/etching process. Other suitable processes are, however, amenable.

(100) As illustrated by the views **2500A-2500D** of FIGS. **25A-25D**, P-type source/drain extensions **124pe** are formed respectively in the mesa **106m** for the P-type planar FET being formed and the fin **106f** for the P-type IO finFET being formed. The P-type source/drain extensions **124pe** are formed, such that polysilicon gate electrodes **108ps** for the P-type planar FET and the P-type IO finFET being formed are each sandwiched between two of the P-type source/drain extensions **124pe**. A process for forming the P-type source/drain extensions **124pe** may, for example, comprise: 1) forming a photoresist mask **2502** masking the fins **106f** and the mesa **106m** for the remaining FETs being formed using photolithography; 2) implanting dopants into the semiconductor substrate **106** using the photoresist mask **2502** and the hard masks **2402** as a mask; and 3) removing the photoresist mask **2502**. Other suitable processes are, however, amenable. For example, the P-type source/drain extensions **124pe** for the P-type planar FET and the P-type source/drain extensions **124pe** for the P-type IO finFET may be formed separately using different photoresist masks. As another example, the P-type source/drain extensions **124pe** for the P-type IO finFET, but not the P-type planar FET, may be omitted.

(101) As illustrated by the views **2600A-2600D** of FIGS. **26A-26D**, N-type source/drain extensions **124ne** are formed respectively in the mesa **106m** for the N-type planar FET being formed and the fin **106f** for the N-type IO finFET being formed. The N-type source/drain extensions **124ne** may, for example, be formed by the same process used for the P-type source/drain extensions **124pe** (see, e.g., FIGS. **25A-25D**), except that a photoresist mask **2602** may instead be used in place of the photoresist mask **2502** (see, e.g., FIGS. **25A-25D**). Other suitable processes are, however, amenable. For example, the N-type source/drain extensions **124ne** for the N-type planar FET and the N-type source/drain extensions **124ne** for the N-type IO finFET may be formed separately using different photoresist masks.

(102) As illustrated by the views **2700A-2700D** of FIGS. **27A-27D**, additional P-type source/drain extensions **124pe** are formed in the fin **106f** for the P-type core finFET being formed. Further, additional N-type source/drain extensions **124ne** are formed in the fin **106f** for the N-type core finFET being formed. Note that the additional N-type source/drain extensions **124ne** are not visible within FIGS. **27A-27D** but may be seen in FIGS. **29A-29D**. The additional P-type source/drain extensions **124pe** may, for example, be formed as illustrated and described at FIGS. **25A-25D** using an appropriately patterned photoresist mask, whereas the N-type source/drain extensions **124ne** may, for example, be formed as illustrated and described at FIGS. **26A-26D** using an appropriately patterned photoresist mask.

(103) Also illustrated by the views **2700A-2700D** of FIGS. **27A-27D**, sidewall spacers **120** are

formed on sidewalls of the polysilicon gate electrodes **108ps**. A process for forming the sidewall spacers **120** may, for example, comprise: 1) depositing a spacer layer; and 2) etching back the spacer layer to remove horizontal segments of the spacer layer but not vertical segments of the spacer layer. Other suitable processes are, however, amenable.

(104) Also illustrated by the views **2700A-2700D** of FIGS. **27A-27D**, an N-type device mask **2702** is formed covering N-type FETs being formed. The N-type device mask **2702** may, for example, be or comprise silicon nitride and/or some other suitable dielectric(s). A process for forming the N-type device mask **2702** may, for example, comprise depositing a mask layer and patterning the mask layer into the N-type device mask **2702** by a photolithography/etching process. Other suitable processes are, however, amenable.

(105) Also illustrated by the views **2700A-2700D** of FIGS. **27A-27D**, the semiconductor substrate **106** is patterned with the N-type device mask **2702** in place to form SSD openings **2704** respectively overlapping with the P-type source/drain extensions **124pe**. The patterning may, for example, be performed by a photolithography/etching process or some other suitable patterning process. The photolithography/etching process may, for example, comprise: forming a photoresist mask with a layout of the SSD openings **2704** using photolithography; etching the semiconductor substrate **106** with both the photoresist mask and the N-type device mask **2702** in place; and removing the photoresist mask. Other suitable processes are, however, amenable. The etching may, for example, be performed by dry etching or some other suitable type of etching. Further, the etching may, for example, be performed with an etchant having a high selectivity (e.g., a high etch rate) for the semiconductor substrate **106** relative to the trench isolation structure **130**, the N-type device mask **2702**, or any combination of the foregoing.

(106) The patterning forms the SSD openings **2704** at the mesas **106m** with different heights and profiles than the SSD openings **2704** at the fins **106f**. For example, the SSD openings **2704** at the mesas **106m** may have a first height $H_{sub.p}$, whereas the SSD openings **2704** at the fins **106f** may have a second height $H_{sub.fin}$ greater than the first height $H_{sub.p}$. The different profiles and heights may, for example, be due to different epitaxial loading during the patterning. Because the mesas **106m** may be larger than the fins **106f** and/or have a larger pitch than the fins **106f**, feature density may be different at and around the mesas **106m** than at and around the fins **106f**. The different feature densities may lead to different etch rates while forming the SSD openings **2704**. As a result of the different etch rates, the SSD openings **2704** at the mesas **106m** may be shallower than the SSD openings **2704** at the fins **106f**. In some embodiments, the SSD openings **2704** at the mesas **106m** also has a smaller volume than the SSD openings **2704** at the fins **106f**.

(107) As illustrated by the views **2800A-2800D** of FIGS. **28A-28D**, compressive epitaxial structures **128c** are formed in the SSD openings **2704** (see, e.g., FIGS. **27A-27D**) for the P-type FETs being formed. The compressive epitaxial structures **128c** may, for example, be or comprise silicon germanium and/or some other suitable material.

(108) The compressive epitaxial structures **128c** compress channel regions for the P-type FETs being formed to increase carrier mobility and hence to improve performance. Further, the compressive epitaxial structures **128c** fill the SSD openings **2704** for the P-type FETs. Therefore, the compressive epitaxial structures **128c** may at least partially take on the heights and/or profiles of corresponding SSD openings **2704**. Because of this, and because the SSD openings **2704** at the mesas **106m** have different heights and profiles than the SSD openings **2704** at the fins **106f**, the compressive epitaxial structures **128c** at the mesas **106m** may have different heights and profiles than the compressive epitaxial structures **128c** at the fins **106f**.

(109) Epitaxial deposition rates may vary based on the heights and/or profiles of the SSD openings **2704**. Because of this, and because the SSD openings **2704** at the mesas **106m** have different heights and profiles than the SSD openings **2704** at the fins **106f**, the compressive epitaxial structures **128c** at the mesas **106m** may have different deposition rates than the compressive epitaxial structures **128c** at the fins **106f**. The different deposition rates may further lead to the

compressive epitaxial structures **128c** at the mesas **106m** having different heights and profiles than the compressive epitaxial structures **128c** at the fins **106f**. Further, when the compressive epitaxial structures **128c** have multiple different materials, the different deposition rates may lead to the compressive epitaxial structures **128c** at the mesas **106m** having different material compositions than the compressive epitaxial structures **128c** at the fins **106f**.

(110) A process for forming the compressive epitaxial structures **128c** may, for example, comprise epitaxially depositing the compressive epitaxial structures **128c**. Other suitable processes are, however, amenable.

(111) As illustrated by the views **2900A**, **2900B** of FIGS. **29A** and **29B**, the N-type device mask **2702** is removed and a P-type device mask **2902** is formed covering P-type FETs being formed. The removal may, for example, be performed by an etch or some other suitable type of removal process. The P-type device mask **2902** may, for example, be or comprise silicon nitride and/or some other suitable dielectric(s). A process for forming the P-type device mask **2902** may, for example, comprise depositing a mask layer and patterning the mask layer into the P-type device mask **2902** by a photolithography/etching process. Other suitable processes are, however, amenable.

(112) Also illustrated by the view **2900A** of FIG. **29A**, tensile epitaxial structures **128t** are formed respectively overlapping with the N-type source/drain extensions **124ne**. The tensile epitaxial structures **128t** may, for example, be or comprise silicon carbide and/or some other suitable material. The tensile epitaxial structures **128t** impose tensile stress on channel regions for the N-type FETs being formed to increase carrier mobility and hence to improve performance. In some embodiments, the tensile epitaxial structures **128t** are formed overlapping with the N-type source/drain extensions **124ne** by a same or similar process as that described and illustrated for the compressive epitaxial structures **128c**. Such a process may, for example, comprise: 1) patterning the semiconductor substrate **106** with the P-type device mask **2902** in place to form additional SSD openings using the process illustrated and described at FIGS. **27A-27D**; and 2) forming the tensile epitaxial structures **128t** in the additional SSD openings using the process illustrated and described at FIGS. **28A-28D**. In other embodiments, the tensile epitaxial structures **128t** are formed by other suitable processes.

(113) The tensile epitaxial structures **128t** at the mesas **106m** may have different heights and profiles than the tensile epitaxial structures **128t** at the fins **106f** for the same reasons discussed with regard to the compressive epitaxial structures **128c** at FIGS. **28A-28D**. Similarly, the tensile epitaxial structures **128t** at the mesas **106m** may have different material compositions and/or volumes than the tensile epitaxial structures **128t** at the fins **106f** for the same reasons discussed with regard to the compressive epitaxial structures **128c** at FIGS. **28A-28D**.

(114) While FIGS. **27A-27D**, **28A-28D**, **29A**, and **29B** illustrate formation of the compressive and tensile epitaxial structures **128c**, **128t**, the compressive and tensile epitaxial structures **128c**, **128t** may not be formed in alternative embodiments of the method being illustrated. Hence, the method may proceed from formation of the sidewall spacers **120** at FIGS. **27A-27D** to the acts at FIGS. **30A** and **30B** (hereafter described) without forming the compressive and tensile epitaxial structures **128c**, **128t**. Further, while the tensile epitaxial structures **128t** are formed after the compressive epitaxial structures **128c**, the tensile epitaxial structures **128t** may be formed before the compressive epitaxial structures **128c** in alternative embodiments.

(115) As illustrated by the views **3000A**, **3000B** of FIGS. **30A** and **30B**, the P-type device mask **2902** is removed. The removal may, for example, be performed by an etch or some other suitable type of removal process.

(116) Also illustrated by the views **3000A**, **3000B** of FIGS. **30A** and **30B**, P-type source/drain regions **124p** are formed respectively in the mesa **106m** for the P-type planar FET being formed and the fins **106f** for the P-type finFETs being formed. The P-type source/drain regions **124p** are formed overlapping respectively with the P-type source/drain extensions **124pe** and respectively

with the compressive epitaxial structures **128c**. A process for forming the P-type source/drain regions **124p** may, for example, comprise: 1) forming a photoresist mask **3002** masking the fins **106f** and the mesa **106m** for the N-type FETs being formed using photolithography; 2) implanting dopants into the semiconductor substrate **106** and the compressive epitaxial structures **128c** using the photoresist mask **3002** and the hard masks **2402** as a mask; and 3) removing the photoresist mask **3002**. Other suitable processes are, however, amenable. For example, the P-type source/drain regions **124p** for the P-type planar FET, the P-type IO finFET, the P-type core finFET, or any combination of the foregoing may be formed separately using different photoresist masks.

(117) As illustrated by the views **3100A**, **3100B** of FIGS. **31A** and **31B**, N-type source/drain regions **124n** are formed respectively in the mesa **106m** for the N-type planar FET being formed and the fins **106f** for the N-type finFETs being formed. The N-type source/drain regions **124n** may, for example, be formed by the same process used for the P-type source/drain regions **124p** (see, e.g., FIGS. **30A** and **30B**), except that a photoresist mask **3102** may instead be used in place of the photoresist mask **3002** (see, e.g., FIGS. **30A** and **30B**). Other suitable processes are, however, amenable. For example, the N-type source/drain regions **124n** for the N-type planar FET, the N-type IO finFET, the N-type core finFET, or any combination of the foregoing may be formed separately using different photoresist masks.

(118) As illustrated by the views **3200A**, **3200B** of FIGS. **32A** and **32B**, a planarization layer **3202** is deposited over the hard masks **2402** and the trench isolation structure **130**. The planarization layer **3202** may, for example, be a bottom antireflective coating (BARC) or some other suitable material. Further, the planarization layer **3202** may, for example, be deposited by spin on coating or some other suitable deposition process.

(119) Because the mesas **106m** have a larger top surface area compared to the fins **106f**, the planarization layer **3202** may accumulate to a greater extent over the mesas **106m** than over the fins **106f**. As such, the planarization layer **3202** may initially have a greater thickness $T_{sub.pl1}$ over the mesas **106m** than over the fins **106f**. Note that the as-deposited profile of the planarization layer **3202** is shown in phantom at the mesas **106m**.

(120) Also illustrated by the views **3200A**, **3200B** of FIGS. **32A** and **32B**, the planarization layer **3202** is thinned at the mesas **106m**, but not the fins **106f**, so the planarization layer **3202** has a thickness $T_{sub.pl2}$ at the mesas **106m** that is about the same as that at the fins **106f**. As seen hereafter, by evening out the thickness of the planarization layer **3202** across the mesas **106m** and the fins **106f**, a subsequent etch to remove the hard masks **2402** is more uniform across the mesas **106m** and the fins **106f**. A process for performing the thinning may, for example, comprise: 1) forming a photoresist mask **3204** masking the fins **106f** using photolithography; 2) etching the planarization layer **3202** with the photoresist mask **3204** in place; and 3) removing the photoresist mask **3204**. Other suitable processes are, however, amenable.

(121) As illustrated by the views **3300A**, **3300B** of FIGS. **33A** and **33B**, a fifth etch is performed into the planarization layer **3202** (see, e.g., FIGS. **32A** and **32B**) and the hard masks **2402** (see, e.g., FIGS. **32A** and **32B**) to remove the hard mask **2402**. Particularly, the fifth etch etches back the planarization layer **3202** until the hard masks **2402** are exposed and then etches back the planarization layer **3202** and the hard masks **2402** in parallel. This persists at least until the hard masks **2402** are removed. The fifth etch may, for example, be a non-selective etch, such that it has substantially the same etch rates for the planarization layer **3202** and the hard masks **2402**.

However, other suitable etches are amenable.

(122) Also illustrated by the views **3300A**, **3300B** of FIGS. **33A** and **33B**, the planarization layer **3202** is removed. The removal may, for example, be performed as part of the fifth etch.

Alternatively, the removal may, for example, be performed independent of and after the fifth etch.

(123) As illustrated by the views **3400A**, **3400B** of FIGS. **34A** and **34B**, top surfaces of the polysilicon gate electrodes **108ps** are co-planarized. Further, a CESL **716** and a first interconnect dielectric layer **708a** are formed between the polysilicon gate electrodes **108ps** and with top

surfaces that are co-planarized with the top surfaces of the polysilicon gate electrodes **108ps**. The first interconnect dielectric layer **708a** may, for example, also be referred to as a zeroth interlayer dielectric (ILD) layer. A process for forming the CESL **716** and the first interconnect dielectric layer **708a**, as well as performing the co-planarization, may, for example, comprise: 1) depositing the CESL **716** over the trench isolation structure **130** and the polysilicon gate electrodes **108ps**; 2) depositing the first interconnect dielectric layer **708a** over the CESL **716**; and 3) performing a planarization into the CESL **716**, the first interconnect dielectric layer **708a**, and the polysilicon gate electrodes **108ps**. Other suitable processes are, however, amenable. The planarization may, for example, be performed by a chemical mechanical polish (CMP) or some other suitable process.

(124) As illustrated by the views **3500A**, **3500B** of FIGS. **35A** and **35B**, the polysilicon gate electrodes **108ps** for the P-type FETs being formed are replaced with P-type gate electrodes **108p** and corresponding portions of a first core dielectric layer **116a**, thereby defining a P-type planar FET **102p**, a P-type IO finFET **104iop**, and a P-type core finFET **104cp**. Further, in the case of the P-type core finFET **104cp**, but not the P-type IO finFET **104iop** and the P-type planar FET **102p**, portions of the second IO dielectric layer **114** directly under the corresponding polysilicon gate electrode **108ps** are removed.

(125) As illustrated by the views **3600A**, **3600B** of FIGS. **36A** and **36B**, the polysilicon gate electrodes **108ps** for the N-type FETs being formed are replaced with N-type gate electrodes **108n** and corresponding portions of a second core dielectric layer **116b**, thereby defining an N-type planar FET **102n**, an N-type IO finFET **104ion**, and an N-type core finFET **104cn**. Further, in the case of the N-type core finFET **104cn**, but not the N-type IO finFET **104ion** and the N-type planar FET **102n**, portions of the second IO dielectric layer **114** directly under the corresponding polysilicon gate electrode **108ps** are removed.

(126) As illustrated by the views **3700A**, **3700B** of FIGS. **37A** and **37B**, hard masks **118** are formed atop the P-type and N-type gate electrodes **108p**, **108n**. The hard masks **118** may be formed by etching back the P-type and N-type gate electrodes **108p**, **108n**, depositing a hard mask layer in the resulting recesses, and performing a planarization into the hard mask layer. Other suitable processes are, however, amenable.

(127) Also illustrated by the views **3700A**, **3700B** of FIGS. **37A** and **37B**, a second interconnect dielectric layer **708b** is formed over the hard masks **118** and the first interconnect dielectric layer **708a**. Further, contact vias **710** are formed extending through the second interconnect dielectric layer **708b** to the FETs. Subsequently wires **712** and interwire vias **714** are formed over and electrically coupled to the contact vias **710** as shown at FIGS. **9A** and **9B**. In some embodiments, a passivation layer **720**, pads **718**, and corresponding pad openings **722** are formed over and electrically coupled to the wires **712** and the interwire vias **714** as illustrated at FIG. **7B**.

(128) While FIGS. **10A** and **10B** through FIGS. **13A** and **13B**, FIGS. **14A-14C** through FIGS. **23A-23C**, FIGS. **24A** and **24B**, FIGS. **25A-25D** through FIGS. **28A-28D**, FIGS. **29A** and **29B** through FIGS. **37A** and **37B** are described with reference to various embodiments of a method, it will be appreciated that the structures shown in FIGS. **10A** and **10B** through FIGS. **13A** and **13B**, FIGS. **14A-14C** through FIGS. **23A-23C**, FIGS. **24A** and **24B**, FIGS. **25A-25D** through FIGS. **28A-28D**, FIGS. **29A** and **29B** through FIGS. **37A** and **37B** are not limited to the method but rather may stand alone separate of the method. While FIGS. **10A** and **10B** through FIGS. **13A** and **13B**, FIGS. **14A-14C** through FIGS. **23A-23C**, FIGS. **24A** and **24B**, FIGS. **25A-25D** through FIGS. **28A-28D**, FIGS. **29A** and **29B** through FIGS. **37A** and **37B** are described as a series of acts, it will be appreciated that the order of the acts may be altered in other embodiments. While FIGS. **10A** and **10B** through FIGS. **13A** and **13B**, FIGS. **14A-14C** through FIGS. **23A-23C**, FIGS. **24A** and **24B**, FIGS. **25A-25D** through FIGS. **28A-28D**, FIGS. **29A** and **29B** through FIGS. **37A** and **37B** illustrate and describe as a specific set of acts, some acts that are illustrated and/or described may be omitted in other embodiments. Further, acts that are not illustrated and/or described may be included in other embodiments.

(129) In some embodiments, replacement of the polysilicon gate electrodes **108ps** at FIGS. **35A** and **35B** and/or FIGS. **36A** and **36B** may be omitted completely or partially in alternative embodiments. For example, the replacement may be omitted for the N-type and P-type planar FETs **102n**, **102p** in some embodiments, such that the polysilicon gate electrodes **108ps** for the N-type and P-type planar FETs **102n**, **102p** may persist into the final structure of the IC.

(130) With reference to FIG. **38**, a block diagram **3800** of some embodiments of a method for forming an IC comprising a planar FET and a finFET is provided.

(131) At **3802**, mesas and fins masked by a first pad layer and a second pad layer are formed. See, for example, FIGS. **10A** and **10B** through FIGS. **12A** and **12B**.

(132) At **3804**, a trench isolation structure is formed surrounding the mesas and the fins. See, for example, FIGS. **13A**, **13B**, and **14A-14C**.

(133) At **3806**, the second pad layer, but not the first pad layer, is removed. See, for example, FIGS. **15A-15C**.

(134) At **3808**, deep wells are formed in a mesa and fins for N-type devices through the first pad layer. See, for example, FIGS. **16A-16C**. In alternative embodiments, this is omitted.

(135) At **3810**, device wells are formed in the mesas, but not the fins, through the first pad layer. See, for example, FIGS. **17A-17C** and FIGS. **18A-18C**.

(136) At **3812**, the first pad layer is removed. See, for example, FIGS. **19A-19C**.

(137) At **3814**, the trench isolation structure is recessed around the mesas, but not the fins, so recessed top surface portions of the trench isolation structure are about even with top surfaces of the mesas. See, for example, FIGS. **20A-20C**. The recessing may, for example, enhance saturation and off current for planar FETs being formed at the mesas. However, the enhancement may be marginal where the separation between top surfaces of the mesas and a top surface of the trench isolation structure is less than about 6 nanometers, 4 nanometers, or some other suitable value. Therefore, in alternative embodiments, the recessing is omitted.

(138) At **3816**, a first IO dielectric layer is deposited covering the fins and the mesas. See, for example, FIGS. **21A-21C**.

(139) At **3818**, device wells are formed in the fins, but not the mesas, through the first IO dielectric layer. See, for example, FIGS. **22A-22C**.

(140) At **3820**, the first IO dielectric layer is removed from the fins, but not the mesas. See, for example, FIGS. **22A-22C**.

(141) At **3822**, the trench isolation structure is recessed around the fins. See, for example, FIGS. **22A-22C**.

(142) At **3824**, a second IO dielectric layer is deposited covering the first IO dielectric layer and the fins. See, for example, FIGS. **23A-23C**.

(143) At **3826**, polysilicon gate electrodes and corresponding hard masks are formed on the mesas and the fins. See, for example, FIGS. **24A** and **24B**.

(144) At **3828**, strained source/drain regions and corresponding source/drain extensions are formed in the mesas and the fins. See, for example, FIGS. **25A-25D** through FIGS. **28A-28D** and FIGS. **29A** and **29B** through FIGS. **31A** and **31B**. In alternative embodiments, the strained source/drain regions do not impose stress on the channels of corresponding channel regions and are therefore not “strained”.

(145) At **3830**, the hard masks are removed. See, for example, FIGS. **32A** and **32B** and FIGS. **33A** and **33B**.

(146) At **3832**, the polysilicon gate electrodes are replaced with metal gate electrodes. See, for example, FIGS. **34A** and **34B** through FIGS. **36A** and **36B**. In alternative embodiments, the metal gate replacement is omitted for some, or all of the polysilicon gate electrodes and the remaining polysilicon gate electrodes persist into the final structure.

(147) While the block diagram **3800** of FIG. **38** is illustrated and described herein as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events is not to be

interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. Further, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein, and one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

(148) In some embodiments, the present disclosure provides an IC including: a semiconductor substrate defining a mesa and a fin; a planar FET overlying the mesa, wherein the planar FET includes a first gate dielectric layer and a first gate electrode, and wherein the first gate electrode overlies the first gate dielectric layer; a finFET overlying the fin, wherein the finFET includes a second gate dielectric layer and a second gate electrode overlying the second gate dielectric layer, and wherein a thickness of the second gate dielectric layer is different than that of the first gate dielectric layer; and an interconnect structure covering and electrically coupled to the planar FET and the finFET, wherein the interconnect structure includes a first pad electrically coupled to the planar FET by wires and vias. In some embodiments, the interconnect structure includes a second pad electrically coupled to the finFET by additional wires and additional vias. In some embodiments, the IC further includes: a trench isolation structure overlying the semiconductor substrate and surrounding the mesa and the fin, wherein a top surface of the trench isolation structure has a recess at the mesa, and wherein the first gate electrode is in the recess. In some embodiments, the top surface of the trench isolation structure has a recessed portion in the recess, and wherein the recessed portion extends in a closed path along a border of the mesa and is about even with a top surface of the fin. In some embodiments, the first gate electrode overlaps opposite sidewalls of the mesa. In some embodiments, the IC further includes: a first dielectric layer and a second dielectric layer that collectively define the first gate dielectric layer, wherein the second dielectric layer, but not the first dielectric layer, defines the second gate dielectric layer. In some embodiments, the second dielectric layer has a same material as the first dielectric layer and further has an elevated concentration of nitrogen compared to the first dielectric layer.

(149) In some embodiments, the present disclosure provides another IC including: a semiconductor substrate defining a mesa and a fin; a first gate electrode overlying the mesa and having a bottommost surface substantially elevated above a top surface of the mesa; a pair of first epitaxial structures recessed into the top surface of the mesa, wherein the first gate electrode is between and borders the first epitaxial structures; a second gate electrode overlying the fin, wherein the second gate electrode straddles the fin and extends along opposite sidewalls of the fin; and a pair of second epitaxial structures recessed into a top surface of the fin, wherein the second gate electrode is between and borders the second epitaxial structures, and wherein the second epitaxial structures have different heights than the first epitaxial structures. In some embodiments, the first epitaxial structures have smaller heights than the second epitaxial structures. In some embodiments, bottom surfaces of the first epitaxial structures are elevated relative to bottom surfaces of the second epitaxial structures. In some embodiments, top surfaces of the first epitaxial structures are recessed relative to top surfaces of the second epitaxial structures. In some embodiments, volumes of the first epitaxial structures are less than volumes of the second epitaxial structures. In some embodiments, the first and second epitaxial structures include silicon carbide or silicon germanium. In some embodiments, the first and second epitaxial structures include a first material and a second material, wherein the first epitaxial structures have a first ratio of the first material to the second material, wherein the second epitaxial structures have a second ratio of the first material to the second material, and wherein the first and second ratios are different.

(150) In some embodiments, the present disclosure provides a method including: patterning a semiconductor substrate to define a mesa and a fin; forming a trench isolation structure overlying the semiconductor substrate and surrounding the mesa and the fin; forming a first gate dielectric layer on the mesa, but not the fin; recessing the trench isolation structure around the fin after the forming the first gate dielectric layer; depositing a second gate dielectric layer overlying the first

gate dielectric layer at the mesa and further overlying the fin; forming a first gate electrode overlying the first and second gate dielectric layers at the mesa and partially defining a planar field-effect transistor; and forming a second gate electrode overlying the second gate dielectric layer at the fin and partially defining a fin field-effect transistor. In some embodiments, the forming of the first gate dielectric layer includes: depositing the first gate dielectric layer overlying the fin and the mesa; and patterning the first gate dielectric layer to remove the first gate dielectric layer from the fin, but not the mesa. In some embodiments, the forming of the first and second gate electrodes includes: depositing a gate electrode layer cover the mesa and the fin over the first and second gate dielectric layers; and performing a selective etch into the gate electrode layer to form the first and second gate electrodes from the gate electrode layer. In some embodiments, the method further includes performing an etch into the mesa and the fin to simultaneously form a pair of first openings and a pair of second openings, wherein the first openings are formed in the mesa with the first gate electrode sandwiched between the first openings, wherein the second openings are formed in the fin with the second gate electrode sandwiched between the second openings, and wherein the first openings have different profiles than the second openings. In some embodiments, the method further includes depositing first epitaxial structures and second epitaxial structures respectively in the first and second openings, wherein sidewall thicknesses of the first epitaxial structures are smaller than sidewall thicknesses of the second epitaxial structures part way through the depositing. In some embodiments, the method further includes recessing a top surface portion of the trench isolation structure at the mesa, but not the fin, until the top surface portion is about even with a top surface of the mesa.

(151) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An integrated circuit (IC), comprising: a semiconductor substrate comprising a mesa and a fin; a first semiconductor device overlying the mesa and comprising a first gate electrode, which is completely elevated relative to a top surface of the mesa; a second semiconductor device overlying the fin and spaced from the first semiconductor device, wherein the second semiconductor device comprises a second gate electrode straddling the fin and extending along sidewalls of the fin; and a trench isolation structure inset into the semiconductor substrate and surrounding the mesa and the fin, wherein the trench isolation structure has a first sidewall, which has a top edge elevated relative to the top surface of the mesa and directly underlies the first gate electrode, wherein the trench isolation structure further has a second sidewall, which has a top edge recessed relative to the top surface of the mesa and directly underlies the second gate electrode, and wherein a width of the first gate electrode increases from a bottom surface of the first gate electrode to a top surface of the first gate electrode.
2. The IC according to claim 1, wherein the first sidewall laterally and directly contacts the mesa, and wherein the second sidewall laterally and directly contacts the fin.
3. The IC according to claim 1, wherein the top edge of the first sidewall is elevated relative to the bottom surface of the first gate electrode.
4. The IC according to claim 1, wherein the first gate electrode has a first thickness directly over the mesa and further has a second thickness directly over the trench isolation structure, and wherein

the second thickness is less than the first thickness.

5. The IC according to claim 1, wherein the first and second gate electrodes have individual top surfaces level with each other, and wherein a thickness of the first gate electrode directly over the mesa is less than a thickness of the second gate electrode directly over the fin.

6. The IC according to claim 1, wherein a top surface of the trench isolation structure has a stepped profile stepping down from a first portion of the top surface of the trench isolation structure that directly underlies the first gate electrode to a second portion of the top surface of the trench isolation structure that directly underlies the second gate electrode.

7. The IC according to claim 6, wherein the first portion of the top surface of the trench isolation structure is edge to edge with the first sidewall, and wherein the second portion of the top surface of the trench isolation structure is edge to edge with the second sidewall.

8. An integrated circuit (IC) comprising: a semiconductor substrate comprising a mesa and a fin spaced from the mesa, wherein the mesa comprises a pair of first doped source/drain regions, and wherein the fin comprises a pair of second doped source/drain regions sharing a height with the pair of first doped source/drain regions; a first gate electrode overlying the mesa and having a bottommost surface substantially elevated above a top surface of the mesa; a pair of first epitaxial structures recessed respectively into the pair of first doped source/drain regions of the mesa, wherein the first gate electrode is between and borders the pair of first epitaxial structures; a second gate electrode overlying the fin, wherein the second gate electrode straddles the fin and extends along opposite sidewalls of the fin; and a pair of second epitaxial structures recessed respectively into the pair of second doped source/drain regions of the fin and spaced from the pair of first epitaxial structures, wherein the second gate electrode is between and borders the pair of second epitaxial structures, and wherein the pair of second epitaxial structures have individual bottom surfaces that are vertically offset from individual bottom surfaces of the pair of first epitaxial structures.

9. The IC according to claim 8, wherein the individual bottom surfaces of the pair of first epitaxial structures are elevated relative to the individual bottom surfaces of the pair of second epitaxial structures.

10. The IC according to claim 8, wherein the pair of second epitaxial structures have individual top surfaces that are vertically offset from individual top surfaces of the pair of first epitaxial structures.

11. The IC according to claim 8, wherein the pair of first epitaxial structures and the pair of second epitaxial structures are spaced from each other in a common cross section.

12. The IC according to claim 11, wherein the pair of the first epitaxial structures have individual areas in the common cross section that are less than individual areas that the pair of second epitaxial structures have in the common cross section.

13. The IC according to claim 8, wherein the pair of first epitaxial structures each comprise a first core epitaxial layer and a first liner epitaxial layer surrounding the first core epitaxial layer, wherein the pair of second epitaxial structures each comprise a second core epitaxial layer and a second liner epitaxial layer surrounding the second core epitaxial layer, and wherein the second liner epitaxial layer has a greater thickness than the first liner epitaxial layer.

14. An integrated circuit (IC), comprising: a semiconductor substrate comprising a mesa and a fin spaced from each other; a planar field-effect transistor (FET) comprising a pair of first source/drain regions and a first gate electrode between the pair of first source/drain regions, wherein the first gate electrode overlies the mesa and is completely elevated relative to a top surface of the mesa; and a finFET comprising a pair of second source/drain regions and a second gate electrode between the pair of second source/drain regions, wherein the second gate electrode overlies the fin and extends along sidewalls of the fin; wherein the mesa has a height directly under the planar FET that is greater than a height of the fin directly under finFET, wherein the pair of first source/drain regions are level with the pair of second source/drain regions, wherein the mesa has a mesa

- sidewall facing towards the fin, wherein the fin has a fin sidewall facing towards the mesa, and wherein the fin sidewall has a bottom edge elevated relative to a bottom edge of the mesa sidewall.
15. The IC according to claim 14, wherein the semiconductor substrate comprises an additional fin between the mesa and the fin, wherein the additional fin has a first sidewall facing the mesa and a second sidewall facing the fin, and wherein a height of the first sidewall is greater than a height of the second sidewall.
16. The IC according to claim 14, wherein a source/drain region of the pair of second source/drain regions has a top surface elevated relative to a source/drain region of the pair of first source/drain regions.
17. The IC according to claim 14, wherein the first gate electrode is separated from the top surface of the mesa by a first gate-dielectric thickness, and wherein the second gate electrode is separated from a top surface of the fin by a second gate-dielectric thickness, which is less than the first gate-dielectric thickness.
18. The IC according to claim 1, wherein a peripheral portion of the first gate electrode overhangs and is spaced from the first sidewall of the trench isolation structure.
19. The IC according to claim 8, wherein the pair of second epitaxial structures share a height greater than a height shared by the pair of first epitaxial structures.
20. The IC according to claim 14, wherein the mesa has an additional mesa sidewall facing away from the fin, and wherein the additional mesa sidewall has a height less than a height of the mesa sidewall.
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